INCH-POUND
MIL-PRF-39005/2F
w/Amendment 2
7 October 2022
SUPERSEDING
MIL-PRF-39005/2F
w/Amendment 1
15 September 2017

PERFORMANCE SPECIFICATION SHEET

# RESISTOR, FIXED, WIREWOUND (ACCURATE), NONESTABLISHED RELIABILITY AND ESTABLISHED RELIABILITY STYLE RBR53

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39005.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for style RBR53, nonestablished reliability, established reliability, accurate, wirewound, fixed resistors (see 6.4).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the style designation, terminal, resistance, resistance tolerance, and product level designator. The PIN is derived in accordance with MIL-PRF-39005 and is in the following form:

RBR53L12601FS

| | | |
|---|---|---|
| RBR53 | - | Style |
| L | - | Terminal |
| 12601 | - | Resistance |
| F | - | Resistance tolerance |
| S | - | Product level designator |

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, ATTN: VAT, Post Office Box 3990, Columbus, Ohio 43218-3990 or by email resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A
FSC 5905
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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

| | | |
|---|---|---|
| MIL-PRF-39005 | - | Resistor, Fixed, Wirewound (Accurate), Nonestablished Reliability and Established Reliability, General Specification for |

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, or in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39005.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1.

3.3 Minimum resistance value and applicable tolerance. The applicable tolerance and minimum resistance value shall be as follows:

| Resistance tolerance (in percent) | Minimum resistance (in Ohms) |
|---|---|
| T (±0.01)<br>Q (±0.02)<br>A (±0.05)<br>B (±0.10) | 10.0 |
| F (±1.0) | 0.10 |

3.4 Maximum resistance value. Maximum resistance values and applicable wire sizes shall be as follows:

| Nominal wire size diameter (in inches) | Absolute diameter (in inches) | Maximum resistance value (megohms) |
|---|---|---|
| 0.001 | 0.0009 | 0.500 |
| 0.0009 | 0.00081 | 0.750 |
| 0.0008 | 0.00072 | 1.1 |

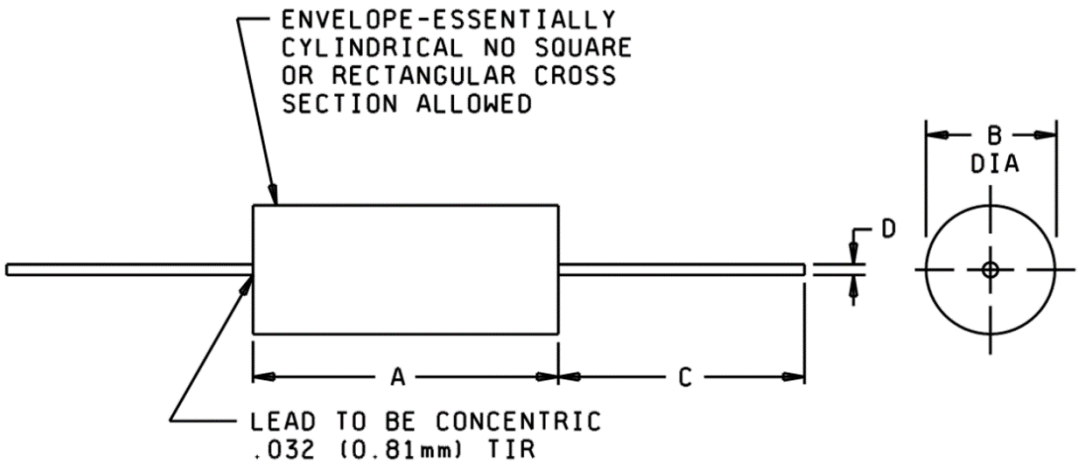
3.5 Power rating. Power rating shall be 0.33 watt.

3.6 Maximum voltage. Maximum voltage shall be 300 volts direct current or peak.

3.7 Maximum weight. The maximum weight shall be 5 grams.



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| Dimension | Inches | Millimeters |
|---|---|---|
| A | 0.750 +0.020, -0.032 | 19.05 +0.51, -0.81 |
| B | 0.375 ±0.015 | 9.52 ±0.38 |
| C | 1.500 min | 38.10 min |
| D | 0.032 ±0.002 | 0.81 ±0.05 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension A is 'clean lead to clean lead'.
4. Resistance measurement points for all values of resistance shall be 0.375 ±0.0625 (9.53 mm ±1.588 mm) from the end of the body.

FIGURE 1. Style RBR53 resistor.

4. VERIFICATION

4.1 Verification. Verification shall be in accordance with MIL-PRF-39005.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-39005 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Unless otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.
c. Packaging requirements (see 5.1).

6.3 Supplementary insulation. Where potential to ground is over 250 volts, supplementary insulation should be provided.

6.4 Suggestive replacement. Due to qualification issues, MIL-PRF-39005/2 does not have a source of supply. Until that time, DLA Land and Maritime drawing 15002 has been created as a suggested replacement.

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Preparing activity:
DLA - CC

Review activities:
Army - AR, AT, CR4, MI
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